Date: 22-Oct-2026

Assg 1 : SQE

Name : Saad Bin Zahid
Reg No. : SP24-BSE-053
Class : BSE-6A

Question : 1. What are the process areas in CMMI level 1 and via which processes are they followed in the IT companies of developed & non-developed countries.

2. Repeat the question 1 for Level 2 & 3

Introduction

Capability Maturity Model Integration (CMMI) is a process improvement model used by software and IT companies to improve product quality, manage projects, reduce risks and deliver on time. It has five maturity levels, from Level 1 (Initial) to Level 5 (Optimizing).

CMMI Level 1 – Initial

• Process Areas :

CMMI Level 1 has no formally defined process areas in the traditional staged CMMI-DEV model. At this level, processes are usually informal, unpredictable and dependent on individual skills.

Date: Day:

Processes Followed In IT Companies

- **Developed Countries:**

Some companies use informal planning, coding, testing and debugging. However, many polished companies follow more mature processes and operate at higher CMMI levels.

- **Non-Developed Countries:**

Some small IT companies and startups rely on verbal instructions, manual testing and individual experience. Processes may not be properly documented, making project outcomes less predictable.

Example:

- A software team starts developing an application without a proper project plan, documented requirements or a fixed testing procedure.

CMMI Level 2 — Managed

→ **Process Areas:**

According to the traditional CMMI-DEV model, Level 2 contains seven process areas:

1- Requirements Management (REQM)
2- Project Planning (PP).

Date:

3. Project Monitoring and Control (PMC)
4. Supplier Agreement Management (SAM)
5. Measurement and Analysis (MA)
6. Process and Product Quality Assurance (PPQA)
7. Configuration Management

→ Processes Followed In IT Companies

• Developed Countries

- Companies commonly use project management software, written requirements, version control systems such as Git, code reviews, quality checks and regular progress meetings.
- These practices support planning, monitoring, and quality assurance.

• Non-Developed Countries

- Many IT companies also use Agile or Scrum, GitHub, issue-tracking tools, documented requirements and manual or automated testing.
- However the level of documentation and process consistency varies between organizations.

→ Example

- Before developing a website, a company documents client requirements, estimates the project budget, assigns tasks, tracks progress,

Date: Day:

test the software, and manages code changes through Git.

CMMI Level 3 — Defined

→ Process Areas :

The traditional CMMI-DEV model contains eleven process areas at Level 3 :

1- Requirements Development (RD)
2- Technical solution (TS)
3- Product Integration (PI)
4- Verification (VER)
5- Validation (VAL)
6- Organizational Process Focus (OPF)
7- Organizational Process Definition (OPD)
8- Organizational Training (OT)
9- Integrated Poject Management (IPM)
10- Risk Management (RSKM)
11- Decision Analysis and Resolution (DAR)

→ Processes Followed In IT Companies :

- **Developed Countries :**
- Organizations establish standard software development procedures, provide employee training, conduct formal risk assessments,

Date: Day:

use defined testing strategies and improve processes through lessons learned.

- **Non-Developed Countries:**
- Companies working toward this maturity level may introduce organization-wide coding standards, reusable development processes, formal training, risk management, and standardized testing.
- Adoption depends on the company's size, resources and quality goals.

→Example:
- A company follows a standard software development lifecycle for every project.
- It documents requirements, design the system, performs code reviews and automated tests, validates the final product with customers and records lessons learned.

COMPARISON OF CMMI LEVELS

Level	Main Focus	Nature of Processes
Level 1 Initial	Completing the work	Informal and unpredictable
Level 2 Managed	Managing individual projects	Planned, monitored and controlled

Date: Day:

Level 3 Defined	Standardizing processes across organization	Documented and consistently followed

→ Conclusions

- CMMI helps IT companies improve software quality, productivity and customer satisfaction.
- Level 1 relies on individual effort.
- Level 2 introduces effective project management.
- Level 3 establishes standard processes, across the organization.
- Both developed and non-developed countries have companies at different maturity levels; the difference depends more on an organization's practices, resources and commitment, than on its country's development status.

→ References

- CMMI Institute - CMMI Levels of Capability and Performance
- CMMI for Development (CMMI-DEV), Version 1.3.
- ISACA - CMMI Performance Solutions.